Assignment 2

Name: _____ Student ID: _____ Marks:_____

1. Solve the differential equation with the boundary conditions

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + by = 0, \quad y(0) = 0, \quad y(5) = 0.$$

Determine the possible values of b. [10]

2. Find the Fourier series representation for $f(x) = x$, where $-\pi \leq x \leq \pi$. [10]